College of the North Atlantic

MA1905 Assignment 4

Due: Monday, November 16th by 11:59 PM.

Upload your completed assignment to the Dropbox.

You must submit your own, original work. Submitting work that is not your own will result in a grade of 0.

Part A (34 marks)

You must show all work. Full marks will not be given for correct final answers if any steps are missing.

1. Let p: The computer is connected to the internet.
   Let q: The antivirus software is running.

   Write each statement using logical symbols. (2 marks)

   a) The computer is connected to the internet and the antivirus software is running.

   b) The computer is not connected to the internet or the antivirus software is running.

2. Given p = true, q = false, and r = true, determine the truth value of each expression.

   Show your evaluation steps. (8 marks)

   a) $p \wedge \sim q$

   b) $\sim p \vee (q \wedge r)$

   c) $(p \vee q) \wedge \sim r$

3. Construct a complete truth table for each of the following. Include intermediate columns. (12 marks)

a) $(p \vee q) \wedge \sim p$

b) $(p \rightarrow q) \wedge (q \rightarrow p)$

c) $(p \wedge \sim q) \vee r$

For algorithm questions, use appropriate pseudocode structure and indentation.

4. A parking garage classifies parking based on the number of hours a customer has parked. Write an algorithm using pseudocode that accepts the number of hours parked. The algorithm should display "Extended Parking" if the number of hours is greater than 3. Otherwise, it should display "Standard Parking". (4 marks)

5. An employee can access a secure system only if they have a valid employee ID AND have completed security training. Write an algorithm using pseudocode that accepts whether the employee has a valid ID and whether they have completed security training. The algorithm should display "Access Granted" if both conditions are satisfied. Otherwise, it should display "Access Denied". (4 marks)

6. A computer repair company offers priority service if a customer has a service contract OR the repair is classified as urgent. Write an algorithm using pseudocode that accepts whether the customer has a service contract and whether the repair is urgent. The algorithm should display "Priority Service" if either condition is satisfied. Otherwise, it should display "Regular Service". (4 marks)

Part B (15 marks)

- Go to the D2L Course Homepage.
- Click Assessments > Quizzes > Assignment 4 Part B.
- Complete the questions within D2L and submit.
- Your calculations for this part of the assignment do not need to be submitted.